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**IMPROVING IMAGE CAPTIONING WITH OBJECT
DETECTION KNOWLEDGE IN REMOTE SENSING
IMAGES**

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LIST OF ABBREVIATIONS

RSIC	Remote Sensing Image Captioning
AI	Artificial Intelligence
RSI	Remote Sensing Image
MLAT	Multilayer Aggregated Transformer
GPU	Graphics Processing Unit
YOLO	You Only Look Once
RSICD	Remote Sensing Image Captioning Dataset
TCE	Truncation Cross Entropy
BLEU	Bilingual Evaluation Understudy
CIDEr	Consensus-based Image Description Evaluation
ROUGE	Recall-Oriented Understudy for Gisting Evaluation

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ABSTRACT

Improving Image Captioning with Object Detection Knowledge in Remote Sensing Images

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Image captioning is a commonly used method for summarizing and extracting details from remote sensing data. However, traditional image captioning models at times struggle with generating accurate descriptions due to varying factors. In this study multiple approaches to incorporate an object detection model into the MLAT architecture is proposed. These methods are implemented and evaluated with their ROUGE, BLEU_4, BLEU_3, BLEU_2, BLEU_1 and CIDEr scoring. The main approach taken in this study with the integration of the object detection model is using a YOLO v8 model trained on the TGRS-HRRSD dataset as a separate feature extractor. These features are then fused with the features extracted from the MLAT encoder's CNN with various methods including concatenation, multiplication, gated fusion, cross modal fusion and spatial attention masking. Among these approaches multiplication performs the best with the BLEU-4 score of 0.3295 compared to the base MLAT model which scored 0.3253 on the same metric.

Keywords: Remote Sensing Image Captioning, Computer Vision, MLAT, YOLO

Uzaktan Algılama Görüntülerinde Nesne Tespiti Bilgisi ile Özetlemelerin Zenginleştirilmesi

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Uzaktan algılama verilerini özetlemek ve detayları çıkarmak için görüntü açıklama yaygın olarak kullanılan bir yöntemdir. Ancak geleneksel görüntü açıklama modelleri birçok faktörden ötürü zaman zaman doğru açıklamalar üretmekte zorlanmaktadır. Bu çalışmada MLAT mimarisine bir obje tespit modelini entegre etmek için çeşitli yöntemler önerilmektedir. Bu yöntemler gerçekleştirilmiş; başarıları ROUGE, BLEU_4, BLEU_3, BLEU_2, BLEU_1 ve CIDEr skorları ile ölçülmüştür. Bu çalışmada benimsenen temel yaklaşım TGRS-HRRSD veri setinde eğitilen bir obje tespit modelinin özellik çıkarıcı olarak kullanılmasıdır. Bu özellikler MLAT'ın şifreleyicisinin içerisindeki CNN'in elde ettiği özelliklerle birlikte ekleme, çarpma, kapılı birleştirme, modeller arası birleştirme ve bölgesel odak maskeleri yöntemleri ile birleştirilmiştir. Bu yaklaşımlar arasında çarpma, 0.3295 BLEU-4 skoru ile 0.3253 skorundaki başlangıç modeline göre en başarılı yaklaşım olmuştur.

Anahtar Kelimeler: Uzaktan Algılama Görüntü Açıklamaları, Bilgisayar Görmesi, MLAT, YOLO

1

Introduction

With the increasing utilization of image captioning in many applications including environmental monitoring, urban planning and disaster response the state of the art models keep pushing new boundaries. The increasing availability of high-resolution satellite and aerial imagery further amplifies the need for automated techniques to describe these images effectively. Image captioning, which generates natural language descriptions of images is a commonly used method for summarizing and extracting details from remote sensing data. An example of image captioning can be seen on Fig. 1.1

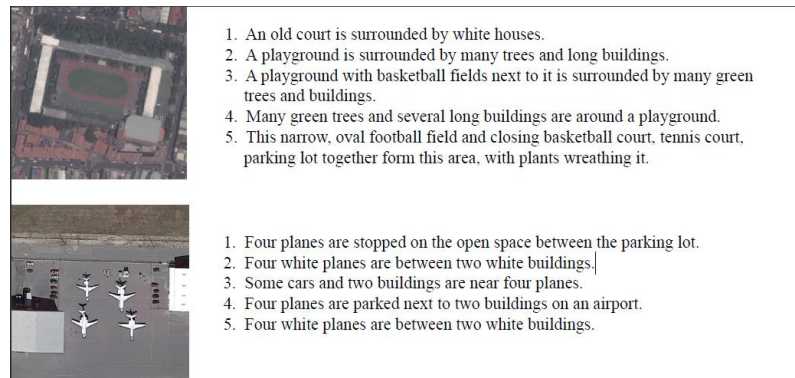


Figure 1.1 RSIC result [1]

However, traditional image captioning models sometimes struggle with creating accurate descriptions due to different factors. Unlike regular images remote sensing images may contain contextual information harder to extract compared to natural images.

To address these challenges this study proposes an approach that combines traditional object detection and image captioning methods. This new method aims to enhance the accuracy of image captioning systems by first identifying key objects in the image using object detection models. An example YOLO object classification result is shown in Fig. 1.2

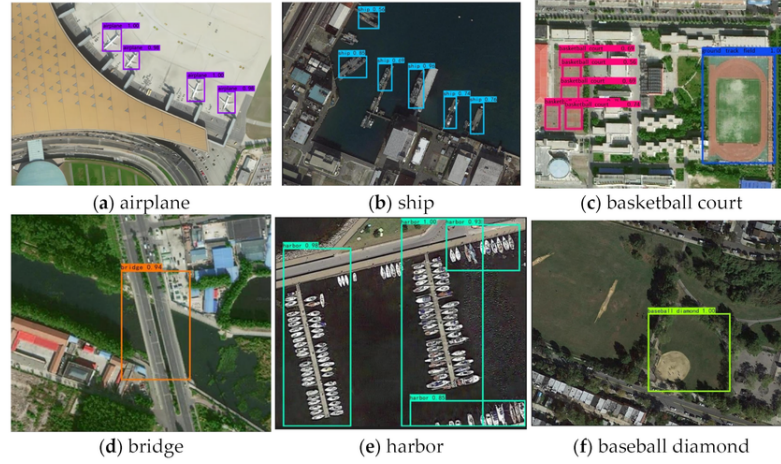


Figure 1.2 Yolo rsic example[2].

1.1 Objective

The primary aim of this study is to incorporate object detection models into the image captioning architecture to further improve accuracy. The goals of the study is as follows:

- Developing an approach that integrates object detection with deep learning based captioning methods.
- Improving caption accuracy to provide further detailed object information.
- Evaluating model performance using commonly used metrics (BLEU, METEOR, CIDEr etc.)

1.2 Preliminary Review

Image captioning has made significant progress in recent years with many state-of-the-art models further improving their architecture. Traditional captioning models have been widely used to generate descriptions of natural images but they present new challenges due to the complexity and higher variability of satellite remote sensing images.

Object detection plays a crucial part in understanding the image before generating textual descriptions. Many state-of-the-art models have been successfully applied to remote sensing tasks, providing detailed identifications of the subjects.

Traditional image captioning models often rely on global features. This study aims to incorporate an object detection model that has been trained for a specific task

into an image captioning model thus further improving it's accuracy and contextual understanding.

2 LITERATURE ANALYSIS

Object detection and remote sensing image captioning are rapidly evolving fields with state-of-the-art approaches improving every day. With many different architectures and techniques being developed it is necessary to understand and analyze current approaches. In this section recent important research on the subject will be covered.

Xuelong Li et al. proposed a new TCE loss for overfitting problem in RSIC dataset caused by the CE loss [3]. To assess the impact of the suggested method, experiments were conducted using three available RSIC datasets: UCM-captions, Sydney-captions, and RSICD.

Zihao Ni et al. proposed a new framework that creates a caption that contains information about the object count for the RSI because typical methods in the field utilize general language, which fail to provide detailed information regarding the number of objects present[4]. Their approach solves the object count problem of the RCIS. To evaluate the accuracy of numerals in the remote sensing images, they developed an evaluation metric named Num-Meteor.

Wang Ke et al. proposed a new model called RN-YOLO based on YOLOv8 to reduce the parameters of the base model[5]. Firstly they incorporated NAM into the feature extraction network to filter the features and then implemented RepGhost_C2f to increase the accuracy while reducing the parameters. Results on the DOTAv1.5, TGRS and RSOD datasets shows that the RN-YOLO approach achieves strong generalization by reducing the parameters by 13.9%.

Chenyang Liu et al. comes up with a new model based on the encoder-decoder approach, as multiscale information is still not efficiently used by current approaches to produce accurate and complete phrases[1]. To extract multiscale information, they fused the features of various ResNet-50 layers in the encoder. They suggested the implementation of the **Multilayer Aggregated Transformer** in the decoder to make adequate use of the retrieved data in order to produce sentences. The structure is

shown in the Figure 4.2.

The MLAT effectively uses features through self attention and aggregation techniques. The BLEU, METEOR, and ROUGE-L scores were calculated for three sentences that were combined in pairs to test sentence diversity. The paper demonstrates that using MLAT as a decoder helps solve the multiscale problem, enabling the model to generate accurate, detailed sentences.

Roberto Castro et al. studied the effects of various hyperparameter configurations on the encoder-decoder architecture regarding the accuracy and BLEU-4 score. Their systems take into consideration the cross-entropy, Kullback-Leibler divergence, mean squared error, negative log-likelihood loss functions and different optimizers. Their experiment showed that cross-entropy with Adam optimizer gives the best result on the BLEU-4 and accuracy score[6].

Ke Zhang et al. created a comprehensive summary of the progress done in the field of RCIS[7]. In their paper they created a detailed review centered around 6 core aspects: learning with auxiliary, few-shot learning, reinforcement learning, attention mechanism, large visual language models and encoder and decoder framework. Furthermore they emphasized the effectiveness of acquiring bottom-up features in transformer-based image captioning.

3.1 System Analysis

Traditional image captioning models often rely on capturing global features which might cause local features to be not fully captured. By incorporating an explicit object detection stage, we aim to improve captioning performance by first identifying key objects in the scene and then using these detections to generate more informative descriptions.

This implementation will be conducted in 3 main steps:

- **Fine-Tuning Object The Detection Model:** Object detection models will be fine-tuned on the dataset. These models will be evaluated with metrics such as ap50, ap50-95 etc.
- **Image Captioning Model:** An image captioning model will be fine-tuned. The captions generated will be evaluated with metrics such as BLEU_4, METEOR, CIDER and ROGUE.
- **Performance Enchantment:** The object detection model obtained will be integrated into the image captioning architecture, further improving its performance.

3.1.1 Environment

The study will be conducted and developed in a CUDA supported Python environment. PYTorch framework will be used with GPU support.

3.1.2 Datasets

In this study publicly available datasets such as **TGRS-HRRSD**[8] and **RSICD**[9] will be used initially.

TGRS-HRRSD is a dataset made by data acquired from Google Earth and Baridu Map. It includes **21,761** images and **55,740** object instances divided into **13** classes. The classes in the dataset **TGRS-HRRSD** are shown in the Table 3.1.

Table 3.1 TGRS-HRRSD class statistics.

Label	Name	N_Train	N_Val	N_Trainval	N_Test	N_All	Mean Resized Scale (pixel)	Resized Scale Std (pixel)
1	ship	950	948	1898	1988	3886	167.44	110.37
2	bridge	1123	1121	2244	2326	4570	246.10	110.53
3	ground track field	859	856	1717	2017	3734	276.50	100.65
4	storage tank	1099	1092	2191	2215	4406	125.60	68.41
5	basketball court	923	920	1843	2033	3876	108.19	57.46
6	tennis court	1043	1040	2083	2212	4295	102.71	38.80
7	airplane	1226	1222	2448	2451	4899	113.21	67.98
8	baseball diamond	1007	1004	2011	2022	4033	231.61	117.85
9	harbor	967	964	1931	1953	3884	163.96	94.16
10	vehicle	1188	1186	2374	2382	4756	41.96	9.99
11	crossroad	903	901	1804	2219	4023	220.54	59.24
12	T junction	1066	1065	2131	2289	4420	198.71	54.88
13	parking lot	1241	1237	2478	2480	4958	122.85	54.45

RSICD is a dataset made up by data acquired from Google Earth, Baidu Map, MapABC, Tianditu. It includes **10921** images with **5 sentence** descriptions per image.

3.1.3 Models

The initial models chosen are **YoloV8**[10] for object detection and **MLAT**[11] for image captioning.

3.1.4 Evaluation

The models will be evaluated using the following metrics:

3.1.4.1 Object Detection

AP50 (Average Precision at IoU=0.50)

Measures the precision of object detection when the predicted bounding box overlaps with the ground truth by at least 50%.

AP50-95

The average of AP calculated at IoU thresholds 0.50-0.95, providing a more comprehensive assessment of detection accuracy.

3.1.4.2 Image Captioning

BLEU-1, 2, 3 and 4

A precision-based metric that evaluates how many 1, 2, 3, or 4 word sequences is in the generated caption which matches in the reference captions.

CIDEr

This metric measures the similarities between generated captions and human written captions using TF-IDF weighting to emphasize important words.

ROUGE

Focuses on the recall of overlapping n-grams, especially useful for evaluating how much of the reference content is captured.

3.2 Feasibility

3.2.1 Technical Feasibility

This study will be done using Python and CUDA supported machine learning frameworks.

3.2.1.1 Hardware Feasibility

Model training will be primarily done on a local computer. The local computers hardware information is shown in the Table 3.2.

Table 3.2 Local hardware.

GPU	NVIDIA GeForce RTX 4060 Max-e
CPU	12th Generation Intel Core i5
RAM	16GB

If more processing power is needed throughout the study Google Colab will be used alongside the local computer.

3.2.2 Legal Feasibility

Every resource used in this study is open for public use.

3.2.3 Economic Feasibility

Frameworks and programming languages used in this study are free and open-source.

3.2.4 Time and Workload Feasibility

Gantt diagram for the study is shown in the Figure 3.1.

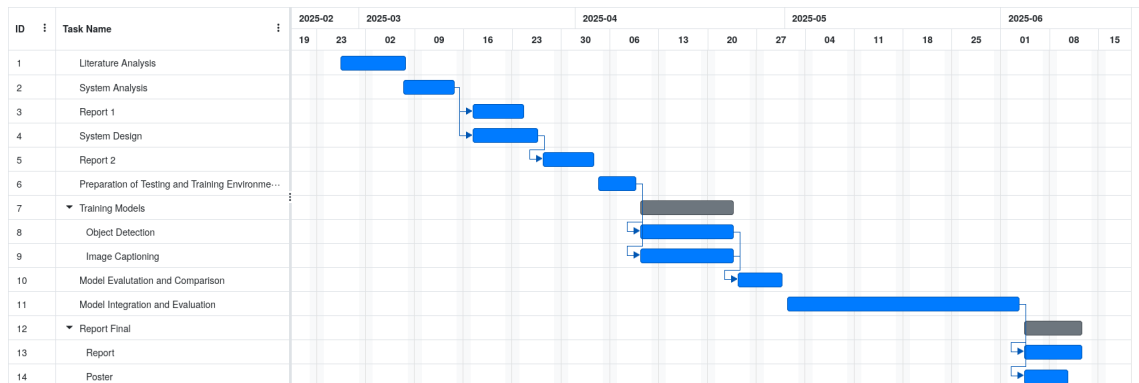


Figure 3.1 Gantt diagram.

4

System Design

4.1 Materials and Dataset

Two different datasets are used throughout this study.

The first dataset is the TGRS-HRRSD dataset for object detection[8]. This dataset contains 21,761 images sourced from Google Earth and Baidu Map. The images contain 55,740 object instances across 13 classes.

Second dataset is RSICD[9]. This dataset is used for the remote sensing image captioning tasks. Remote sensing images are collected from Google Earth, Baidu Map, MapABC, and Tianditu. Although the images have different resolutions, they are standardized at 224x224 pixels. There are a total of 10,921 remote sensing images in the dataset. To every picture, there are five captions describing the picture. This big dataset is popular in RSIC tasks. The sample images illustrate great intraclass variation and low interclass variation, which is why the dataset represents an excellent source of information to be used by researchers. The quantities of different classes in the dataset is shown in Figure 4.1 .

Class	Number	Class	Number	Class	Number
Airport	420	Farmland	370	Playground	1031
Bare Land	310	Forest	250	Pond	420
Baseball Field	276	Industrial	390	Viaduct	420
Beach	400	Meadow	280	Port	389
Bridge	459	Medium Residential	290	Railway Station	260
Center	260	Mountain	340	Resort	290
Church	240	Park	350	River	410
Commercial	350	School	300	Sparse Residential	300
Dense Residential	410	Square	330	Storage Tanks	396
Desert	300	Parking	390	Stadium	290

Figure 4.1 Image Counts of Each Class in RSICD [12]

4.2 Method

In this study a state of the art object detection model is incorporated into the image captioning models to further improve their performance. This is achieved in 3 main steps.

4.2.1 Fine Tuning an Object Detection Model

YoloV8 and YoloV11 models are fine tuned and compared against each other using mAP50val scores. The better performing model is then used to obtain the quantity, location and size information of the detected objects.

4.2.2 Fine Tuning an RSIC Model

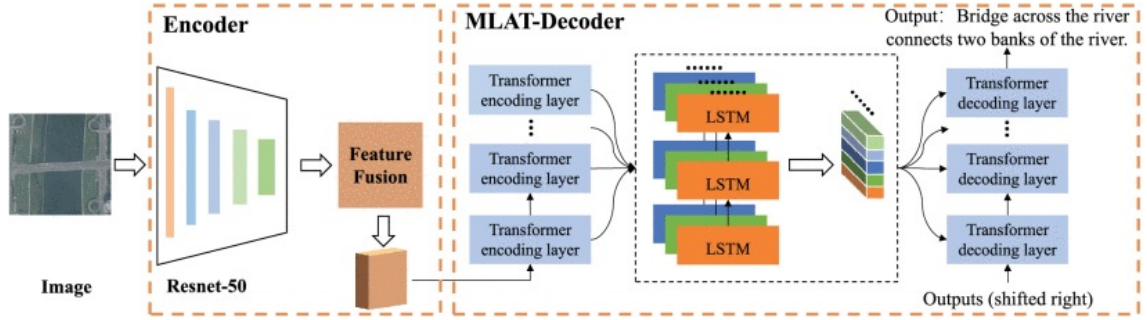


Figure 4.2 MLAT architecture[1].

For RSIC, an image captioning model that adopts an encoder-decoder architecture is used[1]. A MLAT-Decoder is utilized in this model. The model's architecture is shown in Figure 4.2. The encoder extracts multiscale feature vectors by fusing the feature vectors of different layers in ResNet_50 [13]. MLAT makes sufficient use of extracted multispectral properties in the decoder. MLAT improves the links between objects in RSIs and MLAT can precisely describe pictures. In Figure 4.3 an example MLAT output is shown.



1. a base ball field issurrounded by many green trees and many buildings
2. many buildings and green trees are around a base ball field
3. a base ball field issurrounded by many green plants and many buildings

Figure 4.3 Example MLAT output.

4.2.3 Integrating the Models Together

To integrate the object detection model into the RSIC task, the features obtained from the MLAT's encoder is further enriched with features extracted from various layers of a YOLO v8 backbone. To achieve this task the following approaches are suggested:

The first method is to concatenate the features extracted from the last layer of YOLO with the features extracted from RESNet before entering the encoder layer of the MLAT structure.

The second method consists of resizing the features from the last layer of YOLO backbone and the features from RESNet to the same dimensions and then multiplying before entering the encoder layer of the MLAT structure.

Implementation of these first two methods are shown in figure 4.4

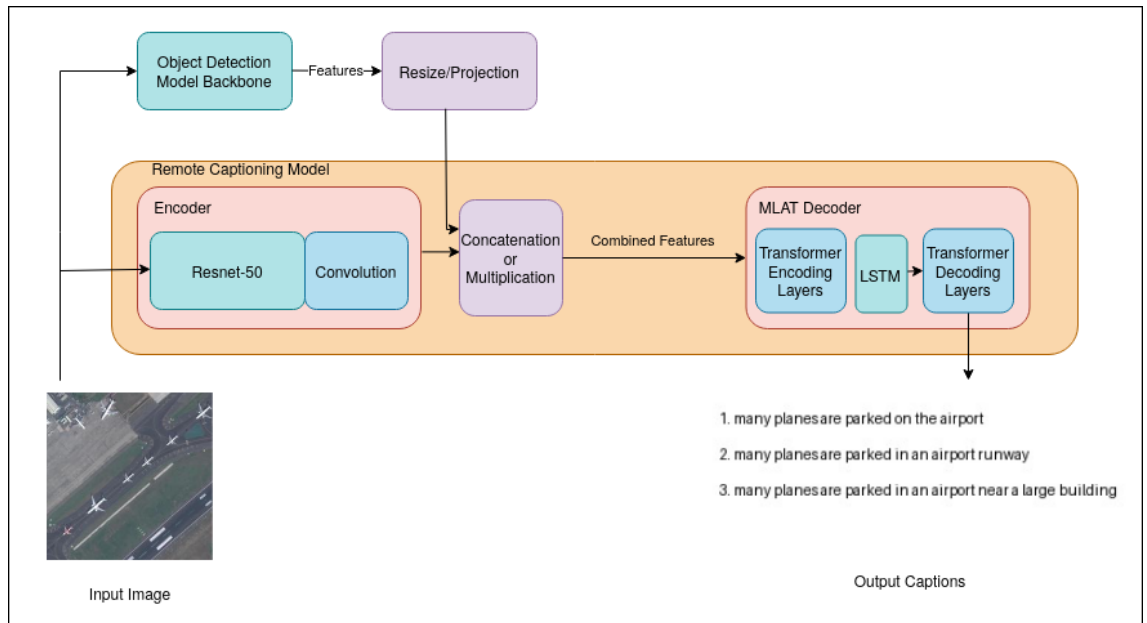


Figure 4.4 Concatenation and multiplication integration.

The third method is the Gate Fusion[14] structure. Rather than directly collecting the features obtained from RESNet and YOLOV8, they are fed into a trainable gating mechanism where how much each source should contribute is dynamically improved. The gate structure can be seen on figure 4.5.

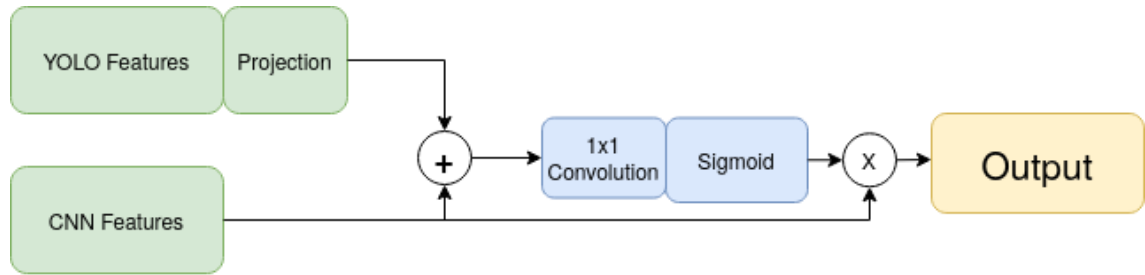


Figure 4.5 Gate fusion implementation.

The fourth method is to fuse the features extracted from a ResNet-50 backbone and object-level features derived from YOLOV8 with cross modal fusion module. The purpose of this module is to align and combine complementary information from both methods before transferring it to the transformer encoder. In addition, normalization is applied to the feature vector (512-DIM) from YOLO.

The fifth method is letting YOLO do full predictions beforehand and implementing an attention mask which causes the RESNet to focus on the regions where YOLO detected relevant objects.

5

EXPERIMENTAL RESULTS

In this section, the results given by the model with different methods are analyzed. To see the effect of the trained YOLO model, untrained YOLO model also used. To measure the performance, BLEU_4, BLEU_3, BLEU_2, BLEU_1, Meteor, ROUGE and CIDEr scores are used.

In the figures 5.1, 5.2, 5.3, 5.4, 5.5, 5.6, 5.7, 5.9 and 5.8 captions generated by the some of the models is shown.



1. many planes are in a terminal on the airport
2. many planes are in a terminal on an airport
3. many planes are parked in a terminal on the airport

Figure 5.1 Base MLAT output.



1. many planes are parked on the airport
2. many planes are parked in an airport runway
3. many planes are parked in an airport near a large building

Figure 5.2 Multiplication output.



1. many planes are in a large building
2. many planes are parked next to a building
3. many planes are parked next to a long building

Figure 5.3 Gate Fusion output.



1. three planes are parked on an airport
2. two planes are parked on an airport
3. three planes are parked in the airport

Figure 5.4 Base MLAT output



1. three planes are parked on an airport
2. four planes are parked on an airport
3. two planes are parked on an airport

Figure 5.5 Multiplication output.



1. three planes are parked on an airport
2. two planes are parked on an airport
3. three planes are parked in the airport

Figure 5.6 Gate Fusion output.



1. two bridges are on a river with many buildings on two sides of it
2. a bridge is on a river with many buildings on two sides of it
3. two bridges are on a river with many buildings on two sides

Figure 5.7 Base output.



1. a bridge is on a river with many green plants on two sides of it
2. two bridges are on a river with many green plants on two sides of it
3. a bridge is on a river with many green plants on two sides

Figure 5.8 Multiplication output.



1. a bridge is on a river with many buildings on two sides of it
2. a bridge is on a river with many buildings on one side of it
3. a bridge is on a river with many green plants on two sides of it

Figure 5.9 Gate Fusion output.

5.1 Performance Analysis

All method results are given in the Table 5.1. The BLEU, ROUGE and CIDEr scores show that the best results are obtained with feature multiplication.

As can be seen from the table 5.1, fusing the outputs of the YOLO model mostly shows no improvement in the model, but it is seen that the multiplication method gives better results than the base model. It also shows that fusing the YOLO model after training it on the TGRS-HRRSD Dataset contributes to the model.

Table 5.1 Performance Metrics

Method	BLEU_1	BLEU_2	BLEU_3	BLEU_4	ROUGE	CIDEr
Base	0.6586	0.4972	0.3953	0.3253	0.4955	0.9270
Concatenation	0.6463	0.4865	0.3890	0.3222	0.4929	0.8638
Attention Mask	0.5157	0.3258	0.2279	0.1688	0.3559	0.4111
Multiplication - Untrained YOLO	0.6460	0.4877	0.3889	0.3214	0.4925	0.9298
Multiplication	0.6584	0.4986	0.3986	0.3295	0.4990	0.9319
Gate Fusion - Untrained YOLO	0.63893	0.4761	0.3773	0.3103	0.4842	0.8478
Gate Fusion	0.6482	0.4864	0.3554	0.3174	0.4901	0.8703
Cross Attention	0.5976	0.4361	0.3409	0.2761	0.4554	0.6986
Cross Attention - Normalized	0.6006	0.4368	0.3337	0.2637	0.4492	0.6896

5.2 Comparative Analysis

As seen in Table 5.1, when BLEU, ROUGE and CIDEr are compared in terms of performance metrics, it cannot be said that there is an approach that improves the MLAT model’s performance by a substantial amount among the applied methods. Although trained on a different dataset, usage of a YOLO model trained on a RSICD shows improvements in final scores.

The Multiplication approach, exceeded the base model in nearly all measures. It achieved the highest BLEU-4 (0.3295) and CIDEr (0.9319), which implies more fluency and relevance in captured generals. This is an indication that the ResNet and YOLO features elementarily interact, enabling the model to effectively fuse complementary spatial and semantic inputs. Nevertheless, it is slightly lower than the base model in BLEU-1 and ROUGE, likely because the attention is being too local.

In comparison, concatenation brought a minor performance drop in terms of all the

metrics, particularly CIDEr (0.8638) and BLEU-2 to BLEU-4. This implies that pure fusion can cause unnecessary or incompatible features to the encoder, thus reduces its efficiency.

The attention mask approach caused significant decrease on the performance of the model. There is a possibility that this decrease is caused by the TGRS-HRRSD and RSICD class differences. When the attention was applied to the area which has a class that in the TGRS-HRRSD but not in the RSICD, the MLAT model might have focused on the parts it could not understand more.

Interestingly, training YOLO backbones (Multiplication and Gated Fusion) is not essential as the untrained versions yield a competitive result in some cases even similar to the trained counterparts. This is evidence that randomly initialized object feature representations may be useful structural bias when incorporated correctly or it could be source of the trained datasets difference.

Lastly, cross-modal fusion and normalized cross-modal fusion performed bad compared to the baseline model, with both having lower BLEU and CIDEr scores. Though this line of approach is theoretically promising, these findings imply that implementing the fusion mechanism may require more sophisticated alignment strategies or extra supervision.

In this study, a YOLO v8 model is integrated into the MLAT RSIC architecture in various ways and its performance is measured with multiple metrics. The YOLO model's backbone was mainly used as a feature extractor and it was used to enrich the features extracted by the CNN encoder of MLAT.

First a YOLO model were trained on a separate dataset. Then a MLAT model were trained normally to be used as a base comparison for our enwer architectures. Then various integration techniques were implemented.

Although most approaches resulted in no or negligible improvement feature multiplication showed some improvement compared to the base MLAT model. The YOLO model's training on a separate remote sensing dataset than the one MLAT was trained on showed some improvement. But this approach created its own challenges too. In the spatial attention approach since the classes YOLO was trying to detect were very different than what the MLAT can create captions with it degraded the performance a substantial amount.

In further studies, based on the findings of this one there are various promising approaches. Firstly, an object detection model that was specifically trained with the same dataset and classes could enhance the performance in some approaches. Secondly instead of trying to use the object detection model as a feature extractor, perhaps its higher level detections can be used in the decoder of the MLAT architecture.

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Project System Informations

System and Software: Ubuntu 24.04, Python, CUDA
Required RAM: 16GB
Required Disk: 32GB